

Exam 3

ECE 2201: Circuit Analysis I

Spring 2025

April 26, 2025

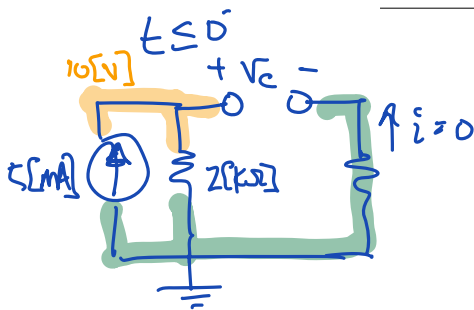
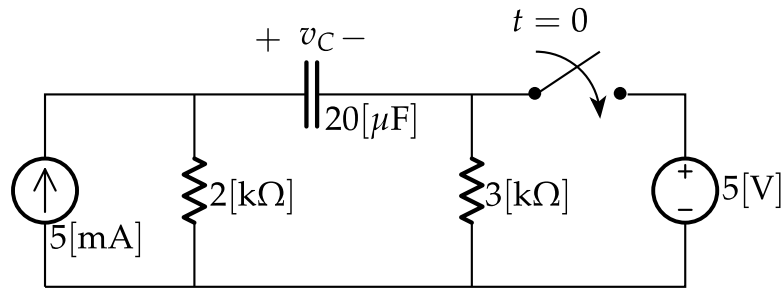
1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem and explain the steps in your solution. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

Name: Solution ID:

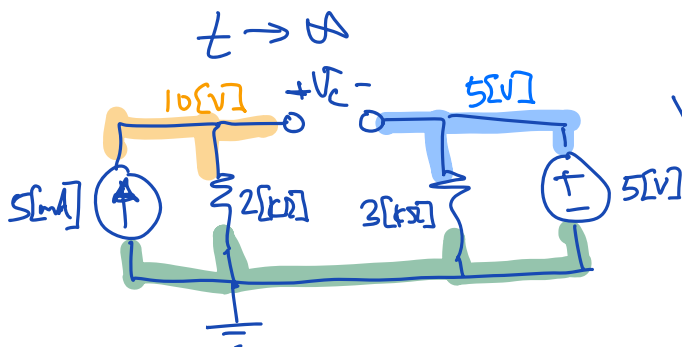
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Problem 1

The switch in the circuit below has been open for a long time before it is closed at $t = 0$. Determine $v_C(0^-)$, $v_C(\infty)$, τ , and $v_C(t)$.

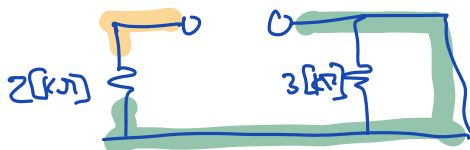


$$v_C(0^-) = 5[\text{mA}] \cdot 2[\text{k}\Omega] = 10[\text{V}]$$



$$v_C(\infty) = 10[\text{V}] - 5[\text{V}] = 5[\text{V}]$$

$$5[\text{V}] + (10[\text{V}] - 5[\text{V}])e^{-t/\tau}$$



$$R_{eq} = 2[\text{k}\Omega]$$

$$\begin{aligned}\tau &= 2[\text{k}\Omega] \cdot 20[\mu\text{F}] \\ &= 40 \times 10^3 \times 10^{-6}[\text{s}] \\ &= 40 \times 10^{-3}[\text{s}] = 40[\text{ms}]\end{aligned}$$

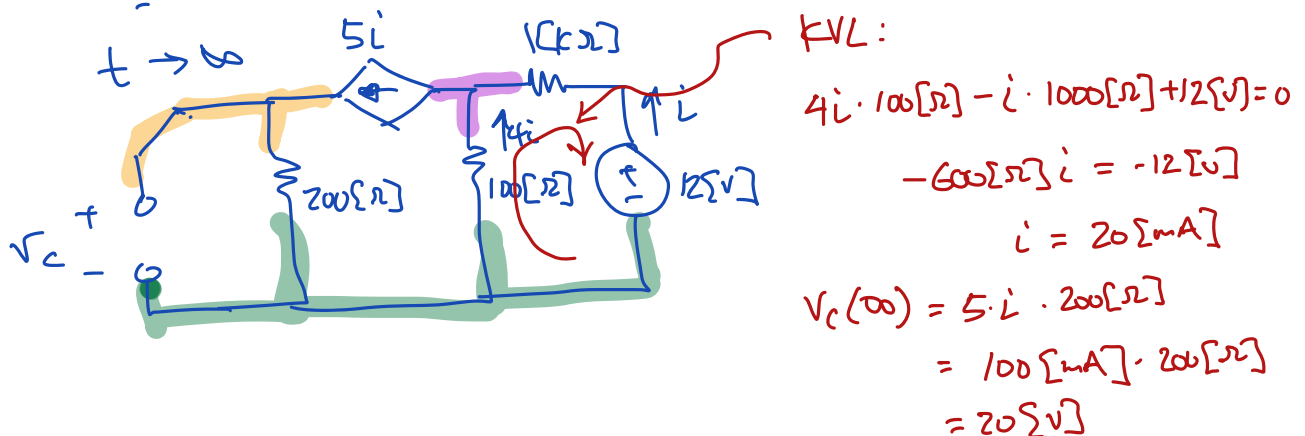
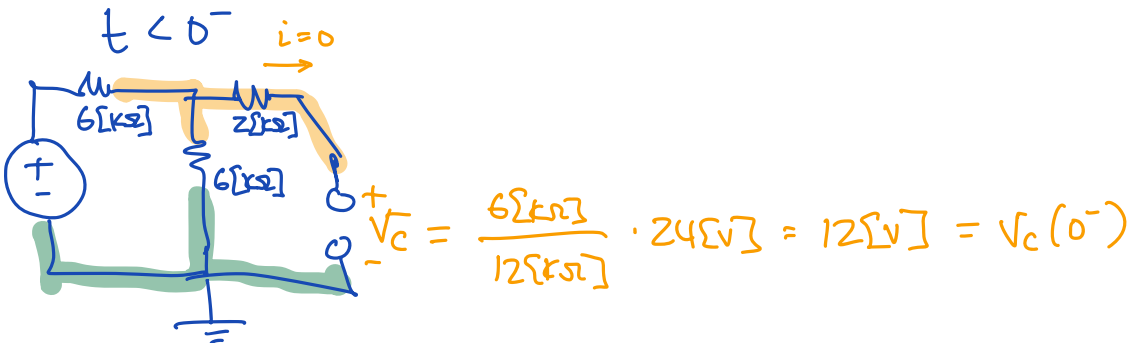
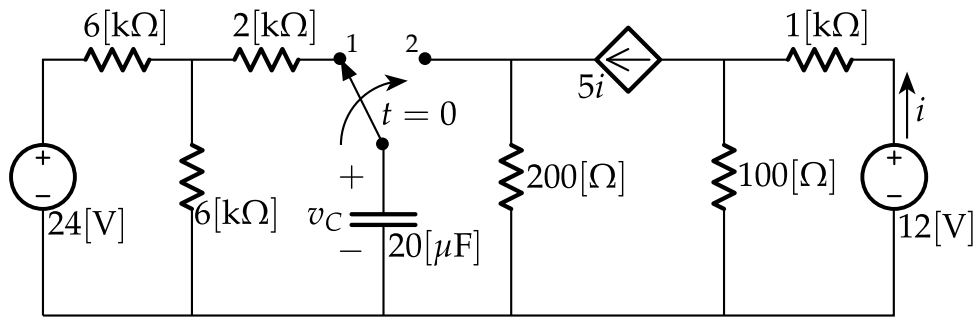
$$v_C(0^-) = 10[\text{V}]$$

$$v_C(\infty) = 5[\text{V}]$$

$$\begin{aligned}\tau &= 40[\text{ms}] \\ v_C(t) &= \left(5 + 5e^{-t/40[\text{ms}]}\right)[\text{V}] \\ &= \left(5 + 5e^{-25t/[\text{ms}]}\right)[\text{V}]\end{aligned}$$

Problem 2

The switch in the circuit below has been in position 1 for a long time before it is moved to position 2 at $t = 0$. Determine $v_C(0^-)$, $v_C(\infty)$, τ (for position 2), and $v_C(t)$.



$$V_{oc} = 5i \cdot 200[\Omega]$$

$$I_{sc} = 5i$$

$$R_{eq} = \frac{V_{oc}}{I_{sc}} = 200[\Omega]$$

$$\tau = 200[\Omega] \cdot 20 \times 10^{-6}[F]$$

$$= 4 \times 10^3 \times 10^{-6}[s]$$

$$= 4[ms]$$

$$v_C(0^-) = \underline{12[V]}$$

$$v_C(\infty) = \underline{20[V]}$$

$$\tau = \underline{4[ms]}$$

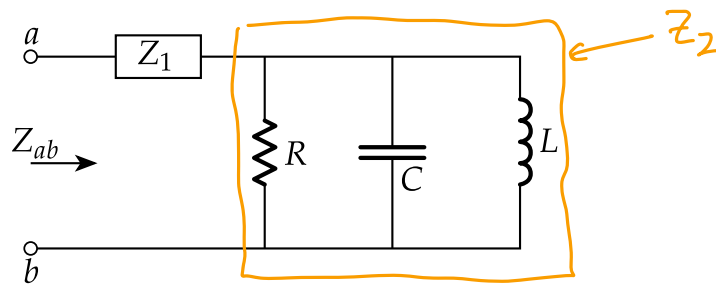
$$v_C(t) = \underline{(20 - 8e^{-t/4[ms]})[V]}$$

$$v_C = 20[V] + (12[V] - 20[V])e^{-t/\tau}$$

Problem 3

The circuit below has been converted to the phasor domain with $R = 5 \text{ [k}\Omega\text{]}$, $Z_C = -j200 \text{ [}\Omega\text{]}$, and $Z_L = j1000 \text{ [}\Omega\text{]}$.

- If Z_1 consists of a single component, find its value if the impedance Z_{ab} is real.
- With $\omega = 1000 \text{ [rad/s]}$, what is the type and value of the component that is Z_1 .



$$Z_2 = \frac{1}{\frac{1}{5000} + \frac{1}{-j200} + \frac{1}{j1000}} \text{ [}\Omega\text{]} = (12.469 - j249.38) \text{ [}\Omega\text{]}$$

$$Z_{ab} = Z_1 + Z_2$$

Since Z_{ab} is real, $Z_1 = -\text{Im}\{Z_2\} = j249.38 \text{ [}\Omega\text{]}$

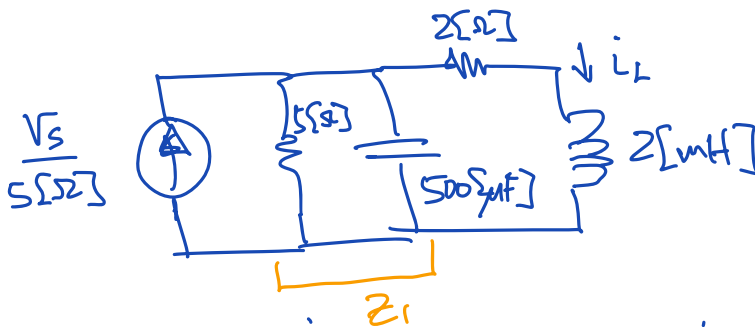
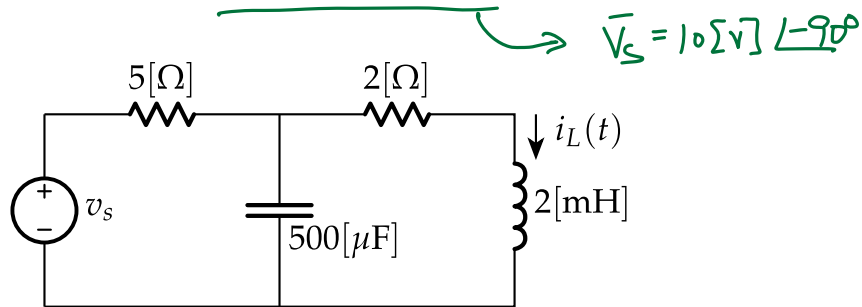
b/c Z_1 is positive, it's an inductor

$$j\omega L = Z_1 \Rightarrow L = \frac{Z_1}{j\omega} = \frac{j249.38 \text{ [}\Omega\text{]}}{j1000} = 249.38 \text{ [mH]}$$

$$Z_1 = \underline{j249 \text{ [}\Omega\text{]}}$$

Type: inductor

Value: 249 [mH]

Problem 4Find $i_L(t)$ in the circuit below given that $v_s = 10 \sin(1000[\text{rad/s}]t)$ [V].

$$\frac{1}{j\omega C} = \frac{-j}{500[\mu\text{F}] \cdot 1000[\frac{\text{rad}}{\text{s}}]} = \frac{-j}{500 \times 10^{-6} \cdot 10^3} = -j2$$

$$j\omega L = j1000[\frac{\text{rad}}{\text{s}}] \cdot 2[\text{mH}] = j10^3 \cdot 2 \times 10^{-3} = j2$$

$$Z_1 = \frac{1}{\frac{1}{5} + \frac{1}{-j2}} [\Omega] = (0.6897 - j1.7241) [\Omega]$$

current divider

$$\bar{I}_L = \frac{\bar{V}_s}{5[\Omega]} \cdot \frac{Z_1}{Z_1 + 2[\Omega] + j2[\Omega]} = \frac{10[\text{V}] \angle -90^\circ}{5[\Omega]} \cdot \frac{Z_1}{Z_1 + (2 + j2)[\Omega]}$$

$$= (-1.32 - j0.38) [\text{A}] = 1.37[\text{A}] \angle -164.1^\circ$$

$$i_L(t) = \underline{1.37[\text{A}]} \cos\left(1000[\frac{\text{rad}}{\text{s}}] \cdot t - 164.1^\circ\right)$$

